

The Supervised Learning Process

Supervised learning involves training a model on a dataset that contains the "right answers" (labels). The model learns from this labeled data and can then make predictions on new, unseen data.

Linear Regression: The First Model

Linear Regression is one of the most widely used machine learning models. Its goal is to fit a **straight line** to a dataset to model the relationship between an input and an output.

- **Example: Housing Price Prediction**

- **Goal:** Predict a house's price based on its size.
 - **Dataset:** A collection of houses with their known sizes (input) and selling prices (output).
 - **Process:**
 1. Plot the known data (size vs. price).
 2. The linear regression model fits the best possible straight line through these data points.
 3. To predict the price of a new house, you find its size on the horizontal axis, trace up to the line, and then across to the vertical axis to get the estimated price.
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Regression vs. Classification (Recap)

- **Regression Model:** Predicts a **continuous numerical value** from an infinite range of possibilities (e.g., predicting a price of \$220,000).
 - **Classification Model:** Predicts a **discrete category** from a small, finite set of possibilities (e.g., cat vs. dog, or disease vs. no disease).
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Key Terminology & Notation

This is the standard notation used in machine learning to describe a training dataset.

- **Training Set:** The dataset used to train the model.
- **Input Variable (Feature):** Denoted by x . This is the input data used for prediction (e.g., size of a house).
- **Output Variable (Target):** Denoted by y . This is the value the model is trying to predict (e.g., price of a house).

- **m**: The total **number of training examples** in the dataset (i.e., the number of rows in the data table).
- (x, y) : Represents a **single training example**—a single input-output pair.
- $(x^{(i)}, y^{(i)})$: Refers to the **i -th training example** in the dataset. The superscript (i) is an index pointing to the i -th row in the table, **it is not an exponent**.
 - For example, $(x^{(1)}, y^{(1)})$ is the first house in the dataset (first row of the data table).